{ Chevy 3er orden
  No Disipativo
  Normalizado

- Con $\varepsilon = 1 \Longrightarrow$ $S_{21} = \dfrac{-0,4913}{s^3 + (0,9883)s^2 + (1,238)s + 0,4913}$ //

$$|S_{21}|^2 = |T|^2 = \frac{-0,2413}{s^6 + (1,5)s^4 + (0,5625)s^2 - 0,2413}$$

$\Rightarrow$ Por Darlington: $|S_{11}|^2 + |S_{21}|^2 = 1$

$$|S_{11}|^2 = 1 - |S_{21}|^2$$

$$= 1 - \frac{-0,2413}{s^6 + (1,5)s^4 + (0,5625)s^2 - 0,2413}$$

$$= \frac{s^6 + (1,5)s^4 + (0,5625)s^2 - 0,2413 + 0,2413}{s^6 + (1,5)s^4 + (0,5625)s^2 - 0,2413}$$

$$|S_{11}|^2 = \frac{\left(s^4 + (1,5)s^2 + (0,5625)\right)s^2}{s^6 + (1,5)s^4 + (0,5625)s^2 - 0,2413}$$

$$S_{11} = \frac{s\left(s^2 + 3/4\right)}{s^3 + (0,9883)s^2 + (1,238)s + 0,4913}$$

- $Z_1 = R_{01} \cdot \frac{S_{11}+1}{S_{11}-1} = R_{01} \cdot \frac{Num + Den}{Num - Den}$

$$= \frac{s^3 + (0,9883)s^2 + (1,238)s + 0,4913 + s^3 + \frac{3}{4}s}{s^3 + (0,9883)s^2 + (1,238)s + 0,4913 - s^3 - \frac{3}{4}s}$$

$$= \frac{2s^3 + (0,9883)s^2 + (1,988)s + 0,4913}{(0,9883)s^2 + (0,488)s + 0,4913}$$

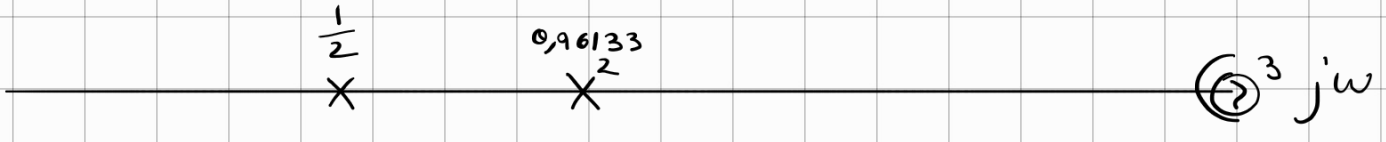
a)

$$Z_1(s) \cong \frac{s^3 + \frac{1}{2}s^2 + s + \frac{1}{4}}{s^2 + \frac{1}{2}s + \frac{1}{2}}$$

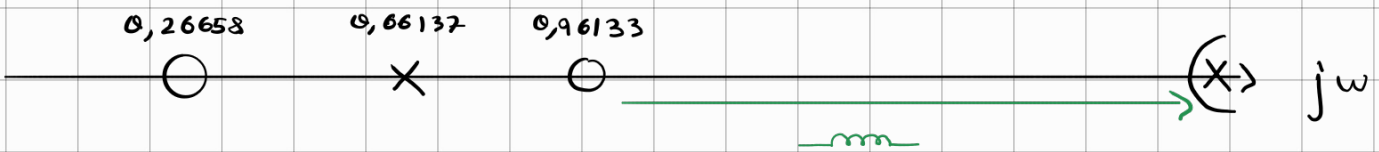
- $Z_1(s) = 2 \cdot \frac{s^3 + \frac{1}{2}s^2 + 1s + \frac{1}{4}}{s^2 + \frac{1}{2}s + \frac{1}{2}}$

$$= \frac{2s^3 + s^2 + 2s + \frac{1}{2}}{s^2 + \frac{1}{2}s + \frac{1}{2}}$$

T:



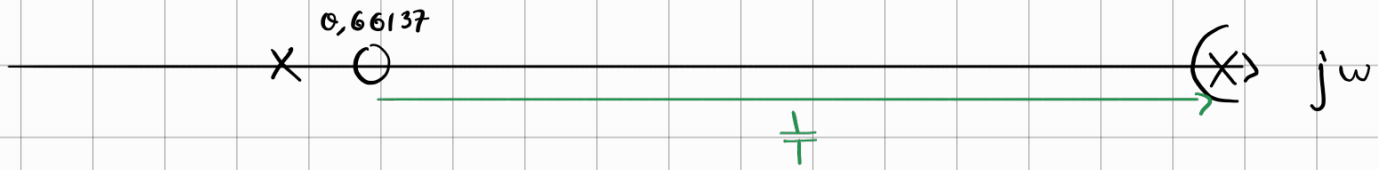
Z_1:



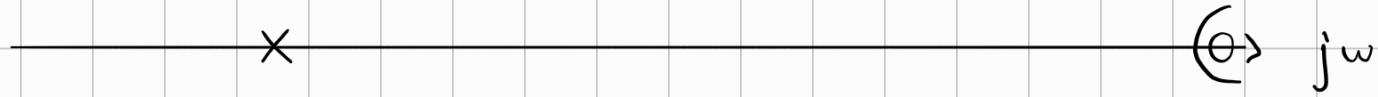
Z_2:



Y_2 :



Y_4 :



Z_4 :



$2s^3 + s^2 + 2s + \frac{1}{2} \;\big|\; s^2 + \frac{1}{2}s + \frac{1}{2}$

$-(2s^3 + s^2 + s) \quad \Big) \; 2s \rightarrow 2$

$s^2 + \frac{1}{2}s + \frac{1}{2} \;\big|\; s + \frac{1}{2}$

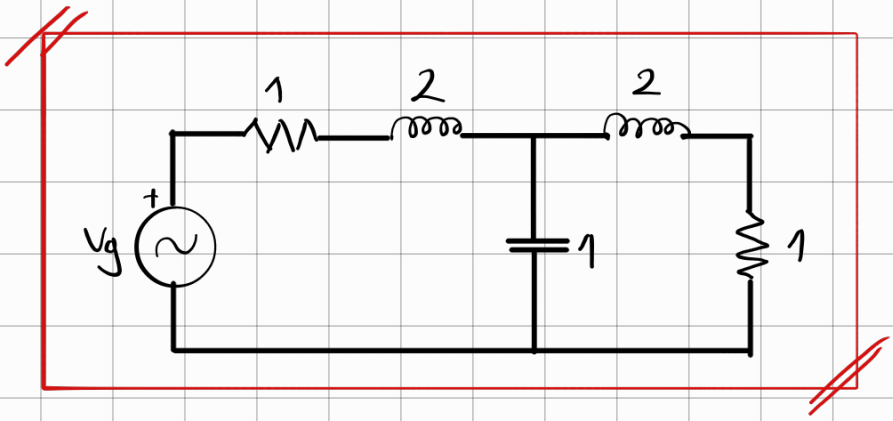
$-(s^2 + \frac{1}{2}s) \quad \Big) \; s \rightarrow \frac{1}{1}$

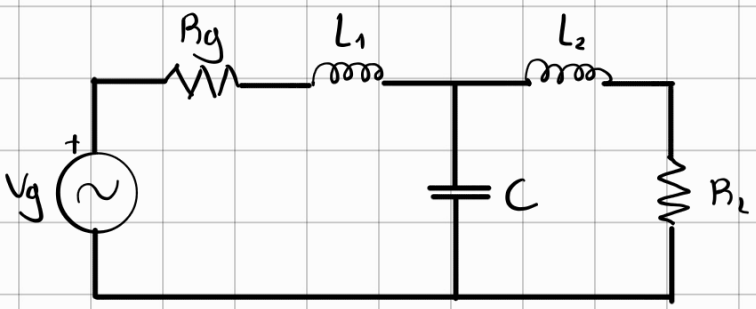
$s + \frac{1}{2} \;\big|\; \frac{1}{2}$

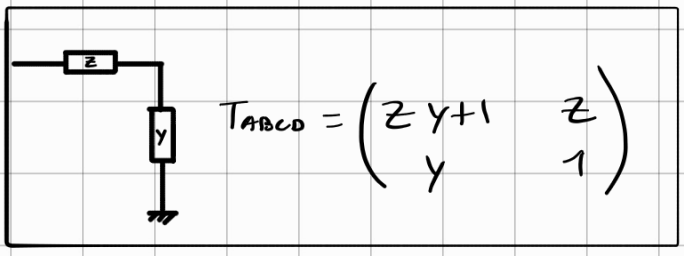
$-s \quad \Big) \; 2s \rightarrow 2$

$\frac{1}{2} \;\big|\; \frac{1}{2}$

$\Big) \; 1 \rightarrow 1$







$$T_{ABCD} = \begin{pmatrix} ZY+1 & Z \\ Y & 1 \end{pmatrix}$$

$$T_{TOT} = \begin{bmatrix} (R_g + sL_1)\cdot sC + 1 & R_g + sL_1 \\ sC & 1 \end{bmatrix} \begin{bmatrix} s\frac{L_2}{R_L} + 1 & sL_2 \\ \frac{1}{R_L} & 1 \end{bmatrix}$$

- $$A = \left(s^2 C L_1 + sCR_g + 1\right)\left(s\frac{L_2}{R_L} + 1\right) + s\frac{L_1}{R_L} + \frac{R_g}{R_L}$$

$$= s^3 \frac{CL_1L_2}{R_L} + s^2\left(CL_2\frac{R_g}{R_L} + CL_1\right) + s\left(CR_g + \frac{L_2}{R_L} + \frac{L_1}{R_L}\right) + \frac{R_g}{R_L} + 1$$

$$= \frac{CL_1L_2}{R_L}\left(s^3 + s^2\left(\frac{R_g}{L_1} + \frac{R_L}{L_2}\right) + s\left(\frac{R_gR_L}{L_1L_2} + \frac{1}{C[L_1 /\!/ L_2]}\right) + \frac{R_g + R_L}{CL_1L_2}\right)$$

$$A^{-1} = \frac{\frac{R_L}{CL_1L_2}}{s^3 + s^2\left(\frac{R_g}{L_1} + \frac{R_L}{L_2}\right) + s\left(\frac{R_gR_L}{L_1L_2} + \frac{1}{C[L_1 /\!/ L_2]}\right) + \frac{R_g + R_L}{CL_1L_2}}$$

- Si $\begin{matrix} m_\omega = \omega_0 \\ m_Z = R_g \end{matrix}$ $\Longrightarrow$

$$\begin{cases} R = R_g \cdot R_N \\ C = \dfrac{C_N}{\omega_0 \cdot R_g} \\ L = \dfrac{R_g}{\omega_0}\cdot L_N \end{cases}$$